VAAL UNIVERSITY
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Module: Business Analysis 3.2 Capstone

Project Title: Stockwise-AI-WMS

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1. Theme Relevance

Our project falls under the theme "AI Solution for Industries". We chose the logistics and warehousing sector. During a site visit to a municipal warehouse in Vereeniging, we saw that staff still use paper logbooks and clipboards to track stock. This makes it impossible to know what is in the warehouse in real-time. Stockwise AI solves this by using machine learning to predict demand, spot theft, and organise the warehouse layout automatically. It fits the theme because it uses modern AI to solve a very real, old-fashioned industry problem.

2. Problem Definition

The main problem is a lack of real-time visibility in municipal warehouses. During our site visit, we observed that stock is recorded manually in logbooks. This manual system causes three major issues. First, it leads to theft and misplacement because there is no way to track items accurately. Second, staff waste hours walking long distances to find items because fast-moving stock is not stored near the dispatch area. Third, there is no automated warning system for stock that is about to expire or run out. Solving this with AI will benefit the Emfuleni Municipality by saving money on lost stock, speeding up service delivery, and giving managers accurate data to make better decisions.

3. Project Stakeholders

Before we built the system, we had to think about who actually cares about this project.

- **The University Side:** Our Lecturer and Moderator will assess our work. We, the group, did the actual research and coding.
- **The Warehouse Side:** The Warehouse Manager will use the dashboard to see stock levels. The Warehouse Clerks will use the chatbot to find items quickly. The Auditors will use the anomaly detection to spot theft. The Clinic Staff are the end-users who will receive stock faster and with less waste.
- **The Technical Side:** Municipal IT Support will maintain the computers, and GitHub is the platform we used to store our code.

4. The 2-Hour Baseline Measurement

During our site visit, we asked the staff how long it takes to find and dispatch a single item. They told us it takes about 2 hours.

We timed it ourselves:

- Looking for the item on the shelf: 45 minutes.
- Walking to get the item: 20 minutes.
- Checking the manual logbook: 30 minutes.
- Taking the item to dispatch and filling in paperwork: 25 minutes.
- This 2-hour baseline is the number we used to measure how much time **Stockwise-AI-WMS** will save the warehouse. With our system, the chatbot tells the worker exactly where the item is, and the dashboard shows the stock level instantly.

5. Business Objectives, Requirements and Risks

Business Objectives: Reduce stock theft by 20%, reduce picking time by 75%, and provide real-time dashboards for managers.

Success 0Criteria: The AI model must accurately predict demand and detect anomalies without too many false alarms.

Risks and Constraints: A risk is that staff may resist using the new system. A constraint is the quality of the data we found, which required heavy cleaning before the AI could use it.

Tools and Techniques: We used Python, Pandas for data cleaning, and Scikit-Learn for the AI models.

6. Theoretical Aspects (Machine Learning Approach)

We used four different techniques for this project:

1. **Time Series Analysis:** We used a Linear Regression model on the last restock date and daily demand to forecast future stock needs.
2. **Anomaly Detection:** We used an Isolation Forest and a Deep Learning Multi-Layer Perceptron (MLP) to detect suspicious picking times. If an item takes unusually long to pick, it might be stolen.

3. **Intelligent Slotting:** We used K-Means Clustering to group items by popularity and demand so fast-moving items go near the front.

4. **Chatbot (NLP):** We built a Natural Language Processing chatbot that looks for keywords to answer staff questions instantly.

7. Model Evaluation

We evaluated our models to ensure accuracy. For our Linear Regression time series model, we checked the R-squared score. For our K-Means clustering model, we calculated the Silhouette Score to measure how well the items were grouped. For our Deep Learning MLP model, we looked at the Reconstruction Mean Squared Error (MSE). Lower MSE means the model is better at spotting anomalies.

8. Practical Solution and Evidence

We built a Python script (*stockwise_ai.py*) that runs all the models and a Streamlit dashboard (*app.py*) for the user interface.

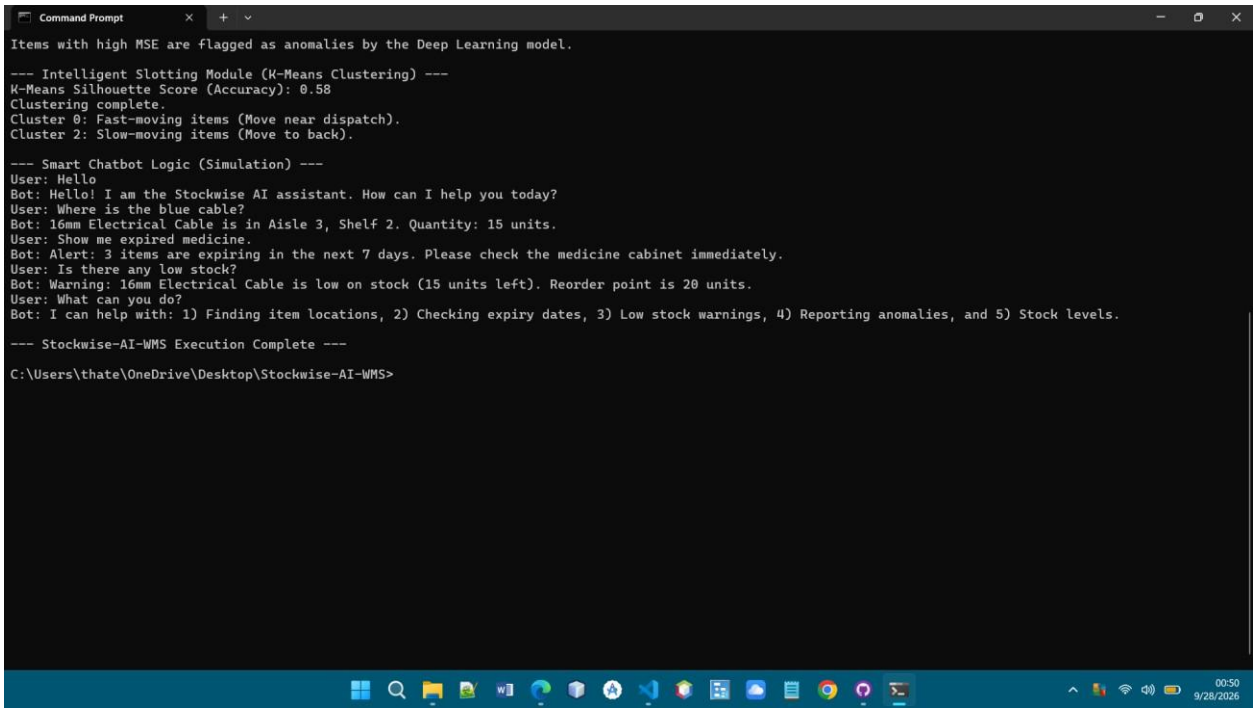
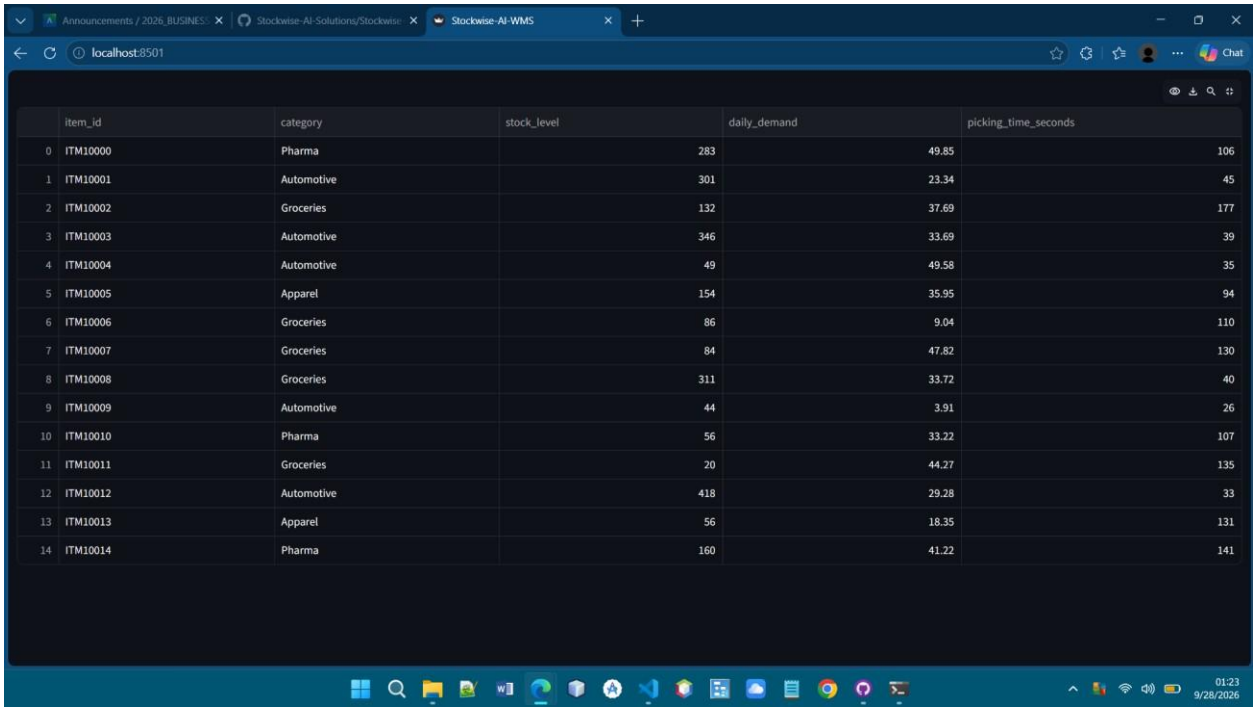


Figure 1: Terminal output showing Time Series forecasting, Deep Learning (MLP) anomaly detection, and K-Means clustering.



	item_id	category	stock_level	daily_demand	picking_time_seconds
0	ITM10000	Pharma	283	49.85	106
1	ITM10001	Automotive	301	23.34	45
2	ITM10002	Groceries	132	37.69	177
3	ITM10003	Automotive	346	33.69	39
4	ITM10004	Automotive	49	49.58	35
5	ITM10005	Apparel	154	35.95	94
6	ITM10006	Groceries	86	9.04	110
7	ITM10007	Groceries	84	47.82	130
8	ITM10008	Groceries	311	33.72	40
9	ITM10009	Automotive	44	3.91	26
10	ITM10010	Pharma	56	33.22	107
11	ITM10011	Groceries	20	44.27	135
12	ITM10012	Automotive	418	29.28	33
13	ITM10013	Apparel	56	18.35	131
14	ITM10014	Pharma	160	41.22	141

Figure 2: Streamlit dashboard showing live stock levels, total items, and low stock alerts.

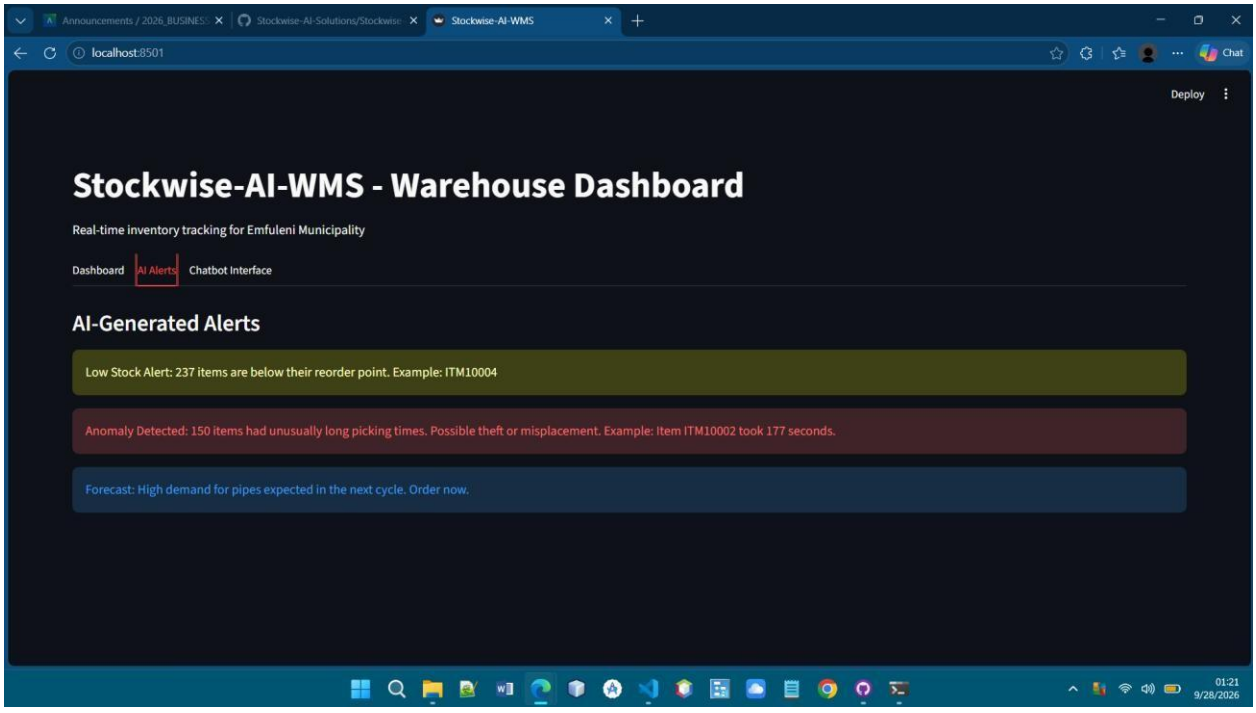


Figure 3: AI detecting low stock and suspicious picking times.

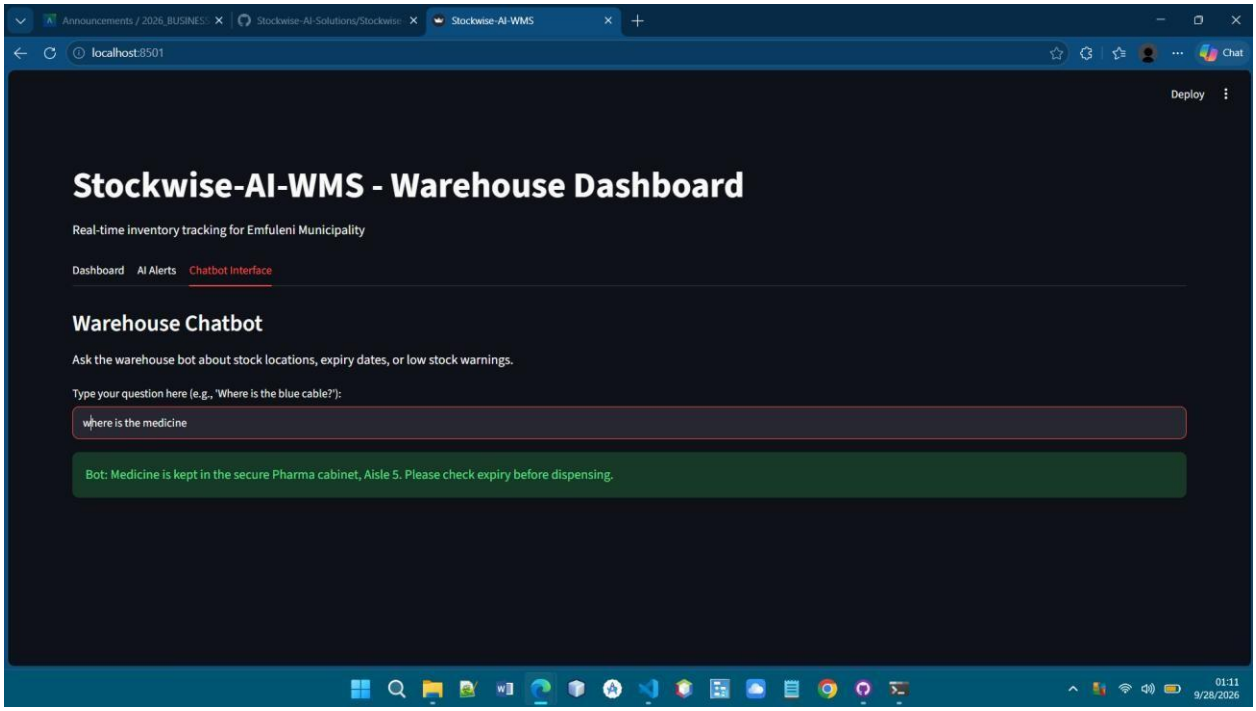


Figure 4: Chatbot interface responding to staff queries using keyword-based NLP.

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10. Appendix: Grammarly Report

AI for Municipal Warehouses

by Thatego Lebidike

General metrics

5,364	839	77	3 min 21 sec	6 min 27 sec
characters	words	sentences	reading time	speaking time

Score



This text scores better than 94% of all texts checked by Grammarly

Writing Issues

11	1	10
Issues left	Critical	Advanced

Plagiarism

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PLAGIARISM DECLARATION

Module: Business Analysis 3.2 Capstone

Project Title: Stockwise-AI-WMS

We, the undersigned, declare that this project is our own original work. All sources used or quoted have been indicated and acknowledged by means of complete references. We declare that this submission is our own work and that it has not been copied from any other student, group, or source.

We understand that plagiarism is a serious offense and that the University will take strict action against it. We confirm that every group member has contributed to this work and agrees to this submission.

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